

RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
IV Semester B.E. Sept/Oct-2024 Examinations
DEPARTMENT: E & C Engineering
COURSE TITLE: MATHEMATICS FOR COMMUNICATION ENGINEERING
(2022 SCHEME)

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

	Q.No	PART-A	Marks
1	1.1	A random process has a sample function of the form $X(t) = \begin{cases} A, & 0 \leq t \leq 1 \\ 0, & \text{Elsewhere} \end{cases}$ Where A is a random variable that is uniformly distributed from 0 to 10. The autocorrelation of the process is _____.	02
	1.2	Let $X(t) = 8 \cos(2\pi t + \theta)$, where θ is a uniformly distributed in the interval $[0, 2\pi]$. The total power of the process = _____ W	02
	1.3	In a binary PCM system, the output signal-to-quantizing-noise ratio is to be held to a minimum of 40 dB. Determine the number of required levels.	02
	1.4	A Speech signal with spectral components limited to frequency band of 500 to 4200Hz is passed through a compander to generate a PCM signal with a sampling rate of 8500samples/sec, if the required output SNR is 32dB then the minimum quantization levels are _____ and the number of bits used to represent each level is _____. (Assume $\mu=255$)	02
	1.5	Number of basis functions required to represent Orthogonal signals = _____ and Antipodal Signals = _____.	02
	1.6	Two devices that exchange data between them have settable clock rates. Case 1 – widely differing rates and Case 2 –Nominally same (but not identical). Multiplexing scheme employed for Case1 = _____ and for Case2 = _____.	02
	1.7	The average energy of the following 4-PAM constellation is = _____. The points are at -3,-1,1 and 3 voltage levels.	02
	1.8	Consider M=2 and M=4 PSK signal constellations. All points have the same transmitted energy E_s. For M=4, determine by how many dB the transmitted signal energy E_s must be increased to achieve the same $d_{\min}$ as M=2.	02
	1.9	The constellation of 16-PSK and 16-QAM is shown in the fig1.9. Which constellation exhibits lower average energy? Justify your answer.	02

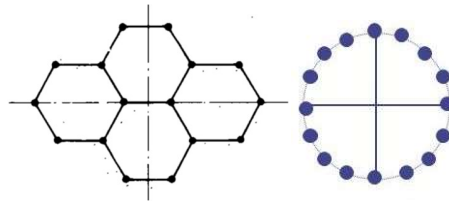
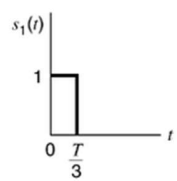
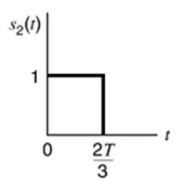
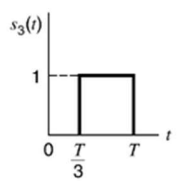
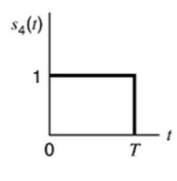


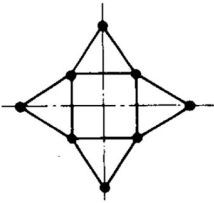
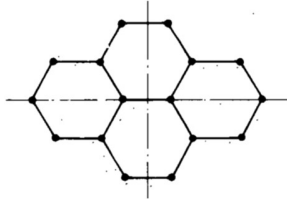
Fig1.9

	1.10	A BPSK signal is used for transmitting information over an AWGN with PSD of $\frac{N_0}{2} = 10^{-10}$ W/Hz. The transmitted signal energy is $E_b = \frac{A^2 T}{2}$, where 'T' is the bit interval and 'A' is the signal amplitude. Determine the signal amplitude _____ (mV) required to achieve an error probability of 10^{-6} , if the data rate is 10kbps.	02
2	a	Given a modulated random signal $Y(t) = A X(t) \sin(\omega_c t + \Theta)$, where $X(t)$ is the random message signal and is a zero-mean stationary random process. The carrier amplitude A and the frequency ω_c are constants and phase Θ is a random variable uniformly distributed over $[-\pi, \pi]$. Assuming that $X(t)$ and Θ are independent, find the mean, autocorrelation and power spectrum of $Y(t)$. Check if $Y(t)$ is a WSS random process.	8
	b	In a random process $X(t) = A \cos(\omega_c t + \Theta)$ where, the amplitude A and the frequency ω_c are constants and phase Θ is a random variable uniformly distributed over $[0, 2\pi]$. Check if $X(t)$ is ergodic in both mean and autocorrelation.	4
	c	A white noise process $X(t)$ is given as input to an RC Low pass filter. Find out (i) The power spectrum at the output of the filter and it's Mean Square value.	4
3	a	State low pass sampling theorem. Deduce the expressions for flat top sampling theorem in time and frequency domain. What is the difference between Impulse sampling and flat top sampling? Illustrate with waveforms.	8
	b	A TV signal has a bandwidth of 4.5MHz. This signal is sampled, quantized and binary coded to obtain a PCM signal. i. Determine the sampling rate if the signal is to be sampled at a rate 20% above the Nyquist rate. ii. If the samples are quantized into 1024 levels, determine the binary pulse rate (bit per sec) of the binary coded signal and the minimum bandwidth required to transmit the signal.	4
	c	Discuss the DPCM scheme with relevant mathematical equations and block diagrams for modulation and demodulation.	4
		OR	
4	a	Deduce an expression for the Quantization noise of a robust quantizer. Check whether the performance of a uniform quantizer with companding is better than that of the uniform quantizer without companding.	10
	b	Consider a DM system designed to accommodate analog message signals limited to a bandwidth of 5KHz. A sinusoidal test signal of amplitude $A=1$ volt and frequency 1KHz is applied to the system. The sampling rate of the system is 50KHz. (i) Calculate the step size required to minimize slope overload. (ii) Calculate the signal to quantization noise ratio of the system for the specified sinusoidal test signal.	06

5	a	Consider the signal $s_1(t)$, $s_2(t)$, $s_3(t)$ and $s_4(t)$ shown in fig 5a . Use Gram Schmidt orthogonalization procedure to find an orthonormal basis for this set of signals. Express these signals in terms of basis functions.     Figure 5a	8M																		
	b	Express the real bandpass signal in terms of its complex lowpass equivalent. Deduce the I/Q modulator and demodulator architectures from the mathematical formulation.	8M																		
OR																					
6	a	The set of 4 signals are described as follows. $S_1(t) = \{A, 0 \leq t < T \quad 0, \text{ Otherwise}$ $S_2(t) = \{A, 0 \leq t < T/2 \quad -A, T/2 \leq t < T$ $S_3(t) = -S_1(t)$ and $S_4(t) = -S_2(t)$ Express the signals in terms of an orthonormal basis functions. Draw the constellation diagram and express the signals in terms of basis functions.	8M																		
	b	Five analog signals S1 to S5 in Table 1 are to be formatted as digital signals for transmission. The allowed overhead includes an 8 bits unique word with 2 bit frame sequence number. Design a multiplexing scheme clearly showing the framing and line rates. What should be the P/D buffer size at the receiver? Table 1 <table><tr><th>Signals</th><th>Bandwidth</th><th>Bits/Sample</th></tr><tr><td>S1</td><td>20 KHz, sampled at 25% above Nyquist rate</td><td>8</td></tr><tr><td>S2</td><td>8KHz, sampled at 20% above Nyquist rate</td><td>10</td></tr><tr><td>S3</td><td>10KHz, sampled at 20% above Nyquist rate</td><td>10</td></tr><tr><td>S4</td><td>15KHz, sampled at 20% above Nyquist rate</td><td>8</td></tr><tr><td>S5</td><td>25K to 30K band pass signal</td><td>10</td></tr></table>	Signals	Bandwidth	Bits/Sample	S1	20 KHz, sampled at 25% above Nyquist rate	8	S2	8KHz, sampled at 20% above Nyquist rate	10	S3	10KHz, sampled at 20% above Nyquist rate	10	S4	15KHz, sampled at 20% above Nyquist rate	8	S5	25K to 30K band pass signal	10	8M
Signals	Bandwidth	Bits/Sample																			
S1	20 KHz, sampled at 25% above Nyquist rate	8																			
S2	8KHz, sampled at 20% above Nyquist rate	10																			
S3	10KHz, sampled at 20% above Nyquist rate	10																			
S4	15KHz, sampled at 20% above Nyquist rate	8																			
S5	25K to 30K band pass signal	10																			

7	a	Deduce the expression for the PSD of a Manchester Line code. Comment on the parameters.	08
	b	Deduce the following for 8-PSK Modulation technique: i. Obtain the basis functions and express the signals in terms of basis functions. Also sketch the constellation. ii. Obtain the minimum Euclidian distance between the symbols and average symbol energy and Draw transmitter and receiver architectures.	08
OR			
8	a	Deduce the following for 8-QAM Modulation technique: i. Obtain the basis functions and express the signals in terms of basis functions. Also sketch the constellation. ii. Obtain the minimum Euclidian distance between the symbols and average symbol energy and Draw transmitter and receiver architectures.	08
	b	Derive SNR of Matched filter. The signals $S_1(t)$ and $S_2(t)$ are defined as	08

		$S_1(t) = \begin{cases} A, & 0 \leq t < T \\ 0, & \text{otherwise} \end{cases}$ $S_2(t) = \begin{cases} A, & 0 \leq t < T/2 \\ -A, & T/2 \leq t \leq T \end{cases}$ i. Determine the impulse response of the matched filter. ii. Plot the output of the matched filter and that of a correlator	
--	--	---	--

9	a	Consider two symbol cases with $P(s_1) > P(s_2)$. Obtain the expression for decision boundary (λ) for which the probability of error is minimum using clock sampling. Using the optimum λ , derive the P_e for two symbol case.	08
	b	Deduce the probability of error expression in terms of the minimum distance between symbols. Obtain the probability of error expression for M-PSK and M-PAM.	08
OR			
10	a	For the 8-QAM signal constellation of fig10a and 16-QAM constellation of fig10b, the distance between any two adjacent points is 'A'. Obtain the average energy for each of the constellations and also obtain optimistic and pessimistic P_e for both the constellations.   Fig10a Fig10b	10
	b	Deduce the expressions of how the estimator computes MAP-based decisions. Draw an optimum receiver architecture for the same.	06

Signature of Scrutinizer:
Name:

Signature of Chairman
Name: